

ENTERPRISE DATA DICTIONARY & BUSINESS SCENARIOS

Global Wealth Banking Customer 360 Architecture Specification

1. Core Banking System Specification (Oracle Core DB)

The core transactional system maintains the definitive, structurally rigid relational records for branches, master profiles, and account ledgers.

A. Branches Table (SRC_BRANCHES)

Field Name	Data Type & Constraints	Business Definition & Governance Rules
BRANCH_ID	NUMBER (PK)	Unique structural identification code for each commercial branch office.
BRANCH_NAME	VARCHAR2(100)	Full registered commercial name of the physical branch location.
REGION	VARCHAR2(50)	Geographic regional classification (e.g., 'CAIRO', 'ALEX', 'DELTA', 'UPPER_EGYPT'). Used for regional macro-performance metrics.

B. Customers Master Table (SRC_CUSTOMERS)

Field Name	Data Type & Constraints	Business Definition & Governance Rules
CUST_ID	NUMBER (PK)	Unique customer identification number. Serves as the primary business anchor key.
CUST_NAME	VARCHAR2(150)	Full legal name of the client. Frequently subject to input syntax errors, trailing spaces, and case disparities at source.
CUST_TYPE	VARCHAR2(20)	Credit segmentation classification tier ('VIP' or 'NORMAL'). Controls downstream credit instrument allocations.
BLACKLIST_FLAG	VARCHAR2(1)	Regulatory sanction and compliance monitoring attribute ('Y' for restricted/banned individuals, 'N' for active status).

C. Accounts Ledger Table (SRC_ACCOUNTS)

Field Name	Data Type & Constraints	Business Definition & Governance Rules
ACC_ID	NUMBER (PK)	Primary operational bank account number across the central ledger.
CUST_ID	NUMBER (FK)	Relational foreign key mapping the account ledger to the definitive master customer record.
BRANCH_ID	NUMBER (FK)	Relational foreign key tracking the account's physical branch origin.
ACC_TYPE	VARCHAR2(30)	Product line type specification ('SAVINGS' or 'CURRENT').
ACC_STATUS	VARCHAR2(20)	Legal operating status of the asset ('ACTIVE' or 'SUSPENDED'). Used to intercept unauthorized card activities.

2. Peripheral System Specification (Incoming Flat Files - CSV)

Peripheral operating logs captured at endpoints or third-party networks, processed as batch data streams.

A. Daily ATM Transactions Log (ATM_Txn_City.csv)

Field Name	Data Type	Business Definition & Operational Constraints
TXN_ID	VARCHAR2(50)	Unique accounting transaction tracking sequence generated by endpoint terminals.
ACC_ID	NUMBER	Target account number processed at the point of interaction. May conflict with suspended states.
TXN_DATE	VARCHAR2(30)	Unstructured transactional timestamp log (exhibits varied string formats 'DD/MM/YYYY' and 'DD-MM-YYYY').
AMOUNT	NUMBER	Net currency volume processed. Prone to mechanical null omissions during connection drops.

B. Annual Card Spending Specifications (CC_Spend_2025_PartX.csv)

Field Name	Data Type	Business Definition & Operational Constraints
CARD_NO	VARCHAR2(30)	Physical primary account number (PAN) of the credit or debit instrument.
CUST_ID	NUMBER	Associated customer identification key. Prone to null omissions due to network parsing errors.
CARD_TYPE	VARCHAR2(20)	Card sub-type designation ('DEBIT' or 'CREDIT'). Utilized to track card threshold compliance.
Q1_SPEND to Q4_SPEND	NUMBER	Pivoted quarterly spending fields summarizing transactional volumes across the financial year.

3. Business Data Integration Scenarios

The following structural paths illustrate the business logic execution across the operational integration pipeline, detailing how anomalous records are routed under core governance directives.

Scenario 1: Standard Flow and Value Stratification (The Happy Path)

Flow Path: A highly compliant retail banking customer holds an active account state within the primary Cairo branch. They maintain exactly one credit card and one debit card instrument. Daily automated ATM transactional data matches perfectly with core customer structures, presenting clean standardized date profiles and unbroken quarterly credit spending sequences across Q1 through Q4.

Business Outcome / Resolution: The integration pipeline successfully synthesizes all incoming channels, dynamically aggregating card spending volumes with baseline ATM cash interactions. It establishes the fiscal third quarter (Q3) as the customer's peak seasonal velocity profile, updating their master financial overview file within the Golden Consolidated Value Report. The profile is prepared for automated tier promotions and targeted high-yield investment marketing.

Scenario 2: The Missing Identity Integration Path (Blind Spot Isolation)

Flow Path: During ingestion of the annual credit card spending batch file, the interface encounters high-value transaction rows containing substantial spending amounts. However, the critical customer identification attribute field (CUST_ID) arrives completely blank (Null) due to an upstream generation malfunction on the external provider network.

Business Outcome / Resolution: To safeguard ledger integrity, the pipeline isolates the row immediately. Because a missing business key prevents standard customer relational lookups, the system computes a physical line trace mapping back to the raw document order (e.g., 'ROW_NO_145'). It bypasses the warehouse and routes the extracted payload to the central Data Audit and Exception Log under the MISSING_PK classification group, enabling manual accounting reconciliations without halting pipeline execution.

Scenario 3: Regulatory Debit Threshold Violations

Flow Path: A retail client is designated within the master database under a standard operating profile (**CUST_TYPE = NORMAL**). Upon executing multi-file batch consolidations across regional card reports, the system uncovers that this specific individual actively operates a debit instrument issued out of the Alexandria regional office, alongside a second distinct debit card registered under the Delta regional branch ledger.

Business Outcome / Resolution: This pattern flags a direct violation of Central Bank regulatory guidelines restricting standard retail tier customers to a single active debit card instrument. The system immediately drops all transaction lines associated with this profile from the core analytical reports. The non-compliant transactional history is expunged from the financial performance ledger and directed to the central Data Audit and Exception Log under the **DEBIT_LIMIT_EXCEEDED** classification group for formal compliance escalation.

Scenario 4: Suspended Accounts and Sanctioned Entities Intervention

Flow Path: A high-volume automated cash withdrawal entry is captured on a physical terminal in the Giza district within the incoming daily ATM transaction log. Upon cross-referencing the account identifier with the central database during real-time transformation lookups, the system detects that the underlying target account is legally frozen (**ACC_STATUS = SUSPENDED**), or that the master client entity is explicitly blacklisted (**BLACKLIST_FLAG = Y**).

Business Outcome / Resolution: In compliance with anti-money laundering (AML) and financial counter-terrorism directives, the pipeline executes an immediate intercept. The transaction is completely expunged from the baseline retail revenue calculations and redirected as a high-risk security breach notification directly to the central Data Audit and Exception Log under the **SUSPENDED_ACCOUNT_TXN** group, triggering automated high-priority alerts for the cybersecurity and internal audit teams.